

**Pseudocode Algoritma Komputasi:
Fitting Resonansi Hamburan Neutron $^{12}\text{C}(n, \text{tot})$ Model
Breit-Wigner Menggunakan Metode Newton-Raphson
Multivariat**

Group H

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1 Algoritma Utama: Pipeline Komputasi

Algorithm 1: MAINPIPELINE – Eksekusi Terintegrasi Tahap 1 hingga Tahap 7

Input : Target nuklida ^{12}C , reaksi (n, tot) , rentang energi $[E_{\min}, E_{\max}] = [1.8, 2.4]$ MeV, toleransi henti $\varepsilon = 10^{-6}$, batas iterasi $k_{\max} = 50$

Output: Parameter optimal $\mathbf{p}^* = [E_r, \Gamma, \sigma_0, \sigma_{bg}]^T$, ketidakpastian $\delta\mathbf{p}$, reduced chi-square χ_{red}^2 , matriks kovariansi $\mathbf{C}$, matriks korelasi $\boldsymbol{\rho}$, plot publikasi

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// Tahap 1: Akuisisi Data, Parsing ASCII, & Pipeline CSV
1 raw_text ← FetchEXFORRaw("C-12", "N,TOT");
2 if raw_text gagal diunduh atau jaringan timeout then
3   raw_text ← FallbackCalibratedDataset();
4  $(E, \sigma, \Delta\sigma) \leftarrow \text{ParseEXFORascii}(\text{raw\_text})$ ;
5 ExportToCSV( $E, \sigma, \Delta\sigma$ , "exfor_c12_resonance.csv");

// Tahap 3: Ekstraksi Otomatis Parameter Awal Fisis
6  $\mathbf{p}_0 \leftarrow \text{EstimateInitialGuess}(E, \sigma)$ ; // Heuristik puncak & FWHM

// Tahap 4: Mesin Solver Newton-Raphson Multivariat
7  $\mathbf{p} \leftarrow \mathbf{p}_0$ ;  $k \leftarrow 0$ ; converged ← False;
8  $\chi_{\text{prev}}^2 \leftarrow \text{ComputeChi2}(\mathbf{p}, E, \sigma, \Delta\sigma)$ ;
9 while  $k < k_{\max}$  dan tidak converged do
    // Tahap 2: Formulasi Analitik Gradien & Matriks Hessian
10   $\mathbf{g} \leftarrow \text{ComputeGradient}(\mathbf{p}, E, \sigma, \Delta\sigma)$ ; // Vektor  $4 \times 1$ 
11   $\mathbf{H} \leftarrow \text{ComputeHessian}(\mathbf{p}, E, \sigma, \Delta\sigma)$ ; // Matriks  $4 \times 4$  Gauss-Newton
12   $\kappa \leftarrow \text{ConditionNumber}(\mathbf{H}) = \|\mathbf{H}\|_2 \cdot \|\mathbf{H}^{-1}\|_2$ ;
13  if  $\kappa > 10^{12}$  atau  $\det(\mathbf{H}) \leq 0$  then
14    catat status ill-conditioned / divergent;;
15    break;
16  Selesaikan sistem linier simultan:  $\mathbf{H} \Delta\mathbf{p} = -\mathbf{g}$  via np.linalg.solve;
17   $\mathbf{p} \leftarrow \mathbf{p} + \Delta\mathbf{p}$ ;
18   $\chi_{\text{curr}}^2 \leftarrow \text{ComputeChi2}(\mathbf{p}, E, \sigma, \Delta\sigma)$ ;
19  Simpan riwayat konvergensi:  $(k, \mathbf{p}, \chi_{\text{curr}}^2, \|\mathbf{g}\|_2, \kappa, \|\Delta\mathbf{p}\|_2)$ ;
20  if  $\|\Delta\mathbf{p}\|_2 < \varepsilon$  atau  $|\chi_{\text{curr}}^2 - \chi_{\text{prev}}^2| < \varepsilon$  then
21    converged ← True;
22   $\chi_{\text{prev}}^2 \leftarrow \chi_{\text{curr}}^2$ ;  $k \leftarrow k + 1$ ;
23  $\mathbf{p}^* \leftarrow \mathbf{p}$ ;  $\mathbf{H}^* \leftarrow \text{ComputeHessian}(\mathbf{p}^*, E, \sigma, \Delta\sigma)$ ;

// Tahap 5: Evaluasi Statistik, Matriks Kovariansi, & Analisis Galat
24  $(\chi_{red}^2, \mathbf{C}, \delta\mathbf{p}, \boldsymbol{\rho}) \leftarrow \text{ComputeStatisticsAndCovariance}(\mathbf{p}^*, \mathbf{H}^*, \chi_{\text{curr}}^2, N)$ ;

// Tahap 6: Uji Ketahanan Numerik (Noise Stress-Test)
25 stress_results ← RunNoiseStressTest( $E, \sigma, \Delta\sigma, \{0.05, 0.15, 0.30\}, 10$ );

// Tahap 7: Visualisasi Saintifik & Validasi Fisis
26 PlotResonanceFitting( $E, \sigma, \Delta\sigma, \mathbf{p}^*, \delta\mathbf{p}$ );
27 PlotConvergenceMetrics(riwayat);
28 PlotNoiseStressTest(stress_results);
29 return  $(\mathbf{p}^*, \delta\mathbf{p}, \chi_{red}^2, \mathbf{C}, \boldsymbol{\rho})$ ;
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2 Sub-Prosedur Rinci per Tahap

2.1 Tahap 2: Model Breit-Wigner & Turunan Analitik

Algorithm 2: BREITWIGNERMODEL & Formulasi Analitik Gradien-Hessian

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1 Function BreitWigner( $E_i, \mathbf{p}$ ):
2    $E_r, \Gamma, \sigma_0, \sigma_{bg} \leftarrow \mathbf{p}$ ;
3    $D_i \leftarrow (E_i - E_r)^2 + (\Gamma/2)^2$ ;
4   return  $\sigma_{bg} + \sigma_0 \frac{(\Gamma/2)^2}{D_i}$ ;

5 Function ComputeGradient( $\mathbf{p}, E, \sigma, \Delta\sigma$ ):
6    $\mathbf{g} \leftarrow [0, 0, 0, 0]^T$ ;
7   for  $i \leftarrow 1$  to  $N$  do
8      $r_i \leftarrow \sigma_i - \text{BreitWigner}(E_i, \mathbf{p}); \quad w_i \leftarrow \frac{1}{(\Delta\sigma_i)^2}$ ;
9      $D_i \leftarrow (E_i - E_r)^2 + (\Gamma/2)^2$ ;
10     $\frac{\partial \sigma_i}{\partial E_r} \leftarrow \frac{2\sigma_0(\Gamma/2)^2(E_i - E_r)}{D_i^2}$ ;
11     $\frac{\partial \sigma_i}{\partial \Gamma} \leftarrow \frac{\sigma_0(\Gamma/2)(E_i - E_r)^2}{D_i^2}$ ;
12     $\frac{\partial \sigma_i}{\partial \sigma_0} \leftarrow \frac{(\Gamma/2)^2}{D_i}$ ;
13     $\frac{\partial \sigma_i}{\partial \sigma_{bg}} \leftarrow 1$ ;
14     $\mathbf{J}_i \leftarrow \left[ \frac{\partial \sigma_i}{\partial E_r}, \frac{\partial \sigma_i}{\partial \Gamma}, \frac{\partial \sigma_i}{\partial \sigma_0}, \frac{\partial \sigma_i}{\partial \sigma_{bg}} \right]^T$ ;
15     $\mathbf{g} \leftarrow \mathbf{g} - 2w_i r_i \mathbf{J}_i$ ;
16  return  $\mathbf{g}$ ;

17 Function ComputeHessian( $\mathbf{p}, E, \sigma, \Delta\sigma$ ):
18   $\mathbf{H} \leftarrow \mathbf{0}_{4 \times 4}$ ;
19  for  $i \leftarrow 1$  to  $N$  do
20     $w_i \leftarrow \frac{1}{(\Delta\sigma_i)^2}$ ;
21     $\mathbf{J}_i \leftarrow \left[ \frac{\partial \sigma_i}{\partial E_r}, \frac{\partial \sigma_i}{\partial \Gamma}, \frac{\partial \sigma_i}{\partial \sigma_0}, 1 \right]^T$ ;
22     $\mathbf{H} \leftarrow \mathbf{H} + 2w_i(\mathbf{J}_i \mathbf{J}_i^T)$ ;           // Aproksimasi Gauss-Newton Kurvatur Kuadratik
23  return  $\mathbf{H}$ ;
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2.2 Tahap 3: Ekstraksi Tebakan Awal Fisis

Algorithm 3: ESTIMATEINITIALGUESS – Heuristik Fisis Bebas Tebakan Acak

Input : Vektor energi E dan penampang lintang σ
Output: Vektor parameter awal $\mathbf{p}_0 = [E_r^{(0)}, \Gamma^{(0)}, \sigma_0^{(0)}, \sigma_{bg}^{(0)}]^T$

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1  $j_{\text{peak}} \leftarrow \arg \max_i (\sigma_i)$ ;
2  $E_r^{(0)} \leftarrow E_{j_{\text{peak}}}$  ; // Lokasi energi puncak resonansi
3  $\sigma_{bg}^{(0)} \leftarrow \min(\sigma_1, \sigma_N)$  ; // Baseline kontinum dari sayap data
4  $\sigma_0^{(0)} \leftarrow \sigma_{j_{\text{peak}}} - \sigma_{bg}^{(0)}$  ; // Tinggi resonansi bersih
5  $\sigma_{\text{half}} \leftarrow \sigma_{bg}^{(0)} + 0.5\sigma_0^{(0)}$  ; // Ambang batas setengah tinggi
6  $E_{\text{above}} \leftarrow \{E_i \mid \sigma_i \geq \sigma_{\text{half}}\}$ ;
7 if  $|E_{\text{above}}| > 1$  then
8    $\Gamma^{(0)} \leftarrow \max(E_{\text{above}}) - \min(E_{\text{above}})$  ; // Estimasi lebar FWHM
9 else
10   $\Gamma^{(0)} \leftarrow 0.05$ ;
11 return  $[E_r^{(0)}, \Gamma^{(0)}, \sigma_0^{(0)}, \sigma_{bg}^{(0)}]^T$ ;

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2.3 Tahap 5: Evaluasi Statistik & Kovariansi

Algorithm 4: COMPUTESTATISTICSANDCOVARIANCE – Matriks Kovariansi & Propagasi Galat

Input : $\mathbf{p}^*$, $\mathbf{H}^*$, $\chi_{\min}^2$, N (jumlah data)
Output: χ_{red}^2 , $\mathbf{C}$, $\delta\mathbf{p}$, $\boldsymbol{\rho}$

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1  $\nu \leftarrow N - 4$  ; // Derajat kebebasan ( $m = 4$  parameter)
2  $\chi_{red}^2 \leftarrow \chi_{\min}^2 / \nu$ ;
3  $\mathbf{C} \leftarrow 2(\mathbf{H}^*)^{-1}$  ; // Inversi matriks kelengkungan Hessian
4 for  $j \leftarrow 1$  to 4 do
5    $\delta p_j \leftarrow \sqrt{C_{jj} \cdot \chi_{red}^2}$  ; // Simpangan baku parameter terbobot
6 for  $j \leftarrow 1$  to 4 do
7   for  $k \leftarrow 1$  to 4 do
8      $\rho_{jk} \leftarrow \frac{C_{jk}}{\sqrt{C_{jj}C_{kk}}}$  ; // Matriks korelasi ternormalisasi
9 return  $(\chi_{red}^2, \mathbf{C}, \delta\mathbf{p}, \boldsymbol{\rho})$ ;

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